FIVE EQUIVALENT TESTS FOR GEOMETRIC REDUCEDNESS VIA PURELY INSEPARABLE EXTENSIONS AND CLOSURES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Embedding finite purely inseparable extensions in the perfect closure. Used at the start of the counted proof (lines 10658-10662). Orders the five conditions so that the implications in one direction are immediate.
- Separable field extensions tested by a characterisation. Invoked near the end of the counted proof (lines 10692-10694). Supplies the separability needed to conclude reducedness of the base change.

Source reference. Original lemma 030W: sources/stacks-algebra/pinned-original-source.tex, source lines 10577–10577. Navigation pointer for \ref{lemma-characterize-separable-field-extensions} used inside the counted span (line 10692); the argument that consumes it is retained in the span.

Source reference. Original lemma 07K2: sources/stacks-algebra/pinned-original-source.tex, source lines 10414–10414. Navigation pointer for \ref{lemma-generic-points-geometrically-reduced} used inside the counted span (line 10693); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00I3: sources/stacks-algebra/pinned-original-source.tex, source lines 10323–10323. Navigation pointer for \ref{lemma-limit-argument} used inside the counted span (line 10669); the argument that consumes it is retained in the span.

Source reference. Original lemma 04KM: sources/stacks-algebra/pinned-original-source.tex, source lines 10179–10179. Navigation pointer for \ref{lemma-make-separably-generated} used inside the counted span (line 10677); the argument that consumes it is retained in the span.

Source reference. Original lemma 00EU: sources/stacks-algebra/pinned-original-source.tex, source lines 4760–4760. Navigation pointer for \ref{lemma-minimal-prime-reduced-ring} used inside the counted span (line 10690); the argument that consumes it is retained in the span.

Source reference. Original lemma 030U: sources/stacks-algebra/pinned-original-source.tex, source lines 10369–10369. Navigation pointer for \ref{lemma-separable-extension-preserves-reducedness} used inside the counted span (line 10679); the argument that consumes it is retained in the span.

Lemma 0.1. *Let k be a field. Let S be a k-algebra. The following are equivalent:*

1. *$k' \otimes_k S$ is reduced for every finite purely inseparable extension k' of k,*
2. *$k^{1/p} \otimes_k S$ is reduced,*
3. *$k^{perf} \otimes_k S$ is reduced, where k^{perf} is the perfect closure of k,*
4. *$\overline{k} \otimes_k S$ is reduced, where $\overline{k}$ is the algebraic closure of k, and*
5. *S is geometrically reduced over k.*

Proof. Note that any finite purely inseparable extension k'/k embeds in k^{perf}. Moreover, $k^{1/p}$ embeds into k^{perf} which embeds into $\overline{k}$. Thus it is clear that (5) $\Rightarrow$ (4) $\Rightarrow$ (3) $\Rightarrow$ (2) and that (3) $\Rightarrow$ (1).

We prove that (1) $\Rightarrow$ (5). Assume $k' \otimes_k S$ is reduced for every finite purely inseparable extension k' of k. Let K/k be an extension of fields. We have to show that $K \otimes_k S$ is reduced. By Lemma 00I3 we reduce to the case where K/k is a finitely generated field extension. Choose a diagram

$$\begin{array}{ccc} K & \longrightarrow & K' \\ \uparrow & & \uparrow \\ k & \longrightarrow & k' \end{array}$$

as in Lemma 04KM. By assumption $k' \otimes_k S$ is reduced. By Lemma 030U it follows that $K' \otimes_k S$ is reduced. Hence we conclude that $K \otimes_k S$ is reduced as desired.

Finally we prove that (2) $\Rightarrow$ (5). Assume $k^{1/p} \otimes_k S$ is reduced. Then S is reduced. Moreover, for each localization $S_{\mathfrak{p}}$ at a minimal prime $\mathfrak{p}$, the ring $k^{1/p} \otimes_k S_{\mathfrak{p}}$ is a localization of $k^{1/p} \otimes_k S$ hence is reduced. But $S_{\mathfrak{p}}$ is a field by Lemma 00EU, hence $S_{\mathfrak{p}}$ is geometrically reduced by Lemma 030W. It follows from Lemma 07K2 that S is geometrically reduced. $\square$